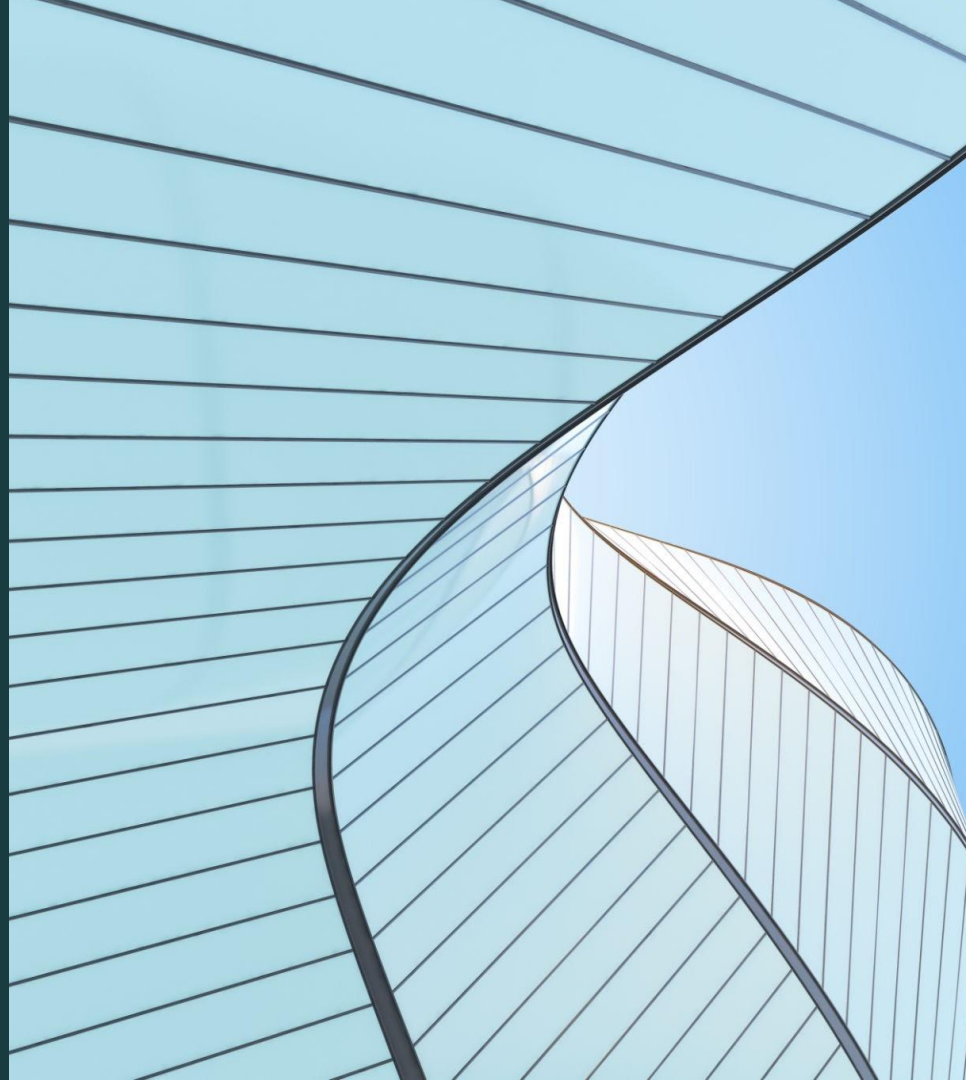


Intro to Python: Movie Ranking Recap

By: Shelsy, Gianella, Jieun, Leslie



Our data

- Two datasets we used :
 - Top 100 popular movies from 2003 and 2022, Rotten Tomatoes movie rating from Kaggle.
- Analyzed how movie trends have changed overall
- Identify any correlations presented through specific features
- Designed a modeling approach to determine whether a movie is likely to be successful or not.

Our data

- **Top 100 popular movies:**
 - 2,000 samples
 - 13 features (e.g. title, year, runtime, certificate, directors, stars, filming location, income, budget)
- **Rotten tomatoes movie rating:**
 - 15,386 samples
 - 13 features(e.g. title, rating, tomatometer rating, audience rating)
- We attempted to merge the two datasets, but due to the large difference in sample sizes, we decided to use them separately.

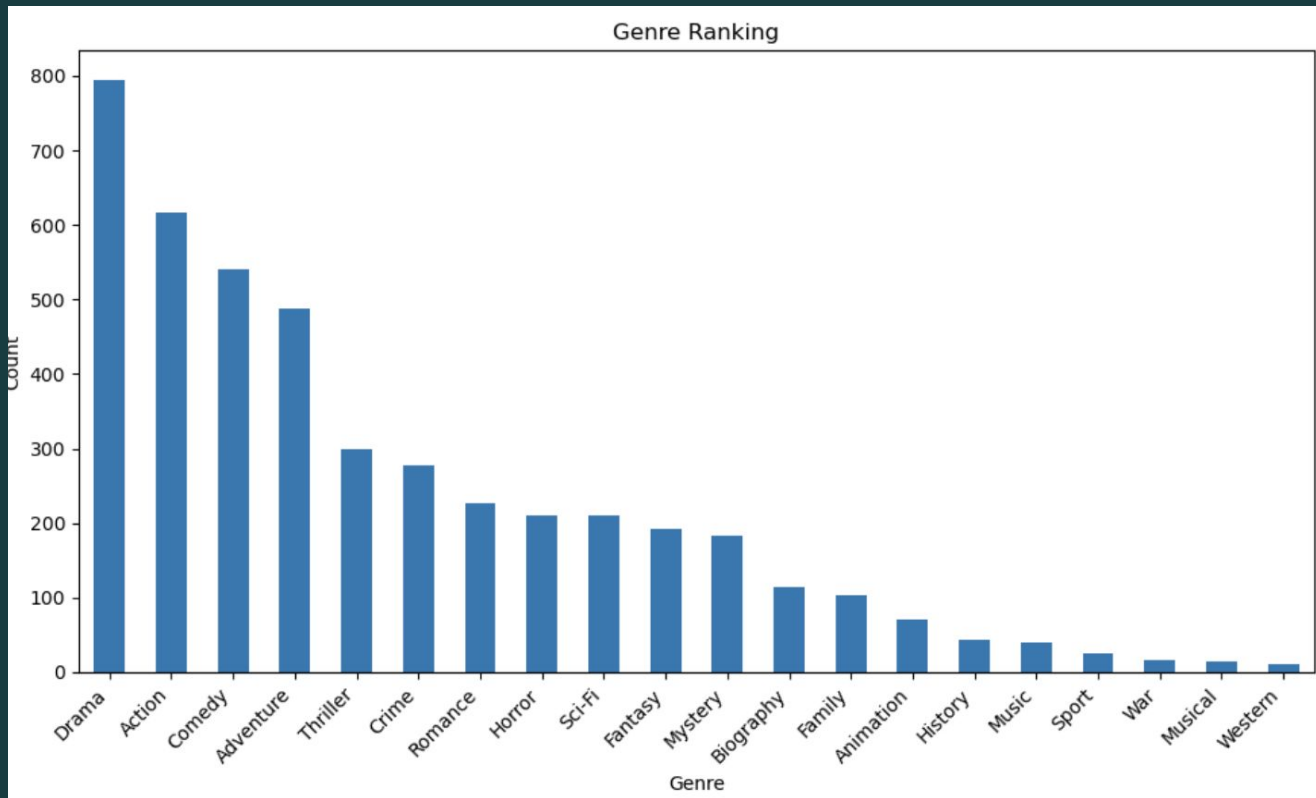
Data cleaning

- Values labeled as 'Unknown' were converted to missing values.
- Since missing values accounted for less than 5% of the total data, we simply dropped them.
- Budget and Income were originally stored as object types, so they were converted to floats. Because the values were in different currencies, currency symbols and whitespace were removed and all amounts were converted to USD.
- ROI(Return on Investment) was added to measure investment efficiency beyond absolute Budget and Income values.

$$\text{ROI (\%)} = \frac{\text{Income} - \text{Budget}}{\text{Budget}} \times 100$$

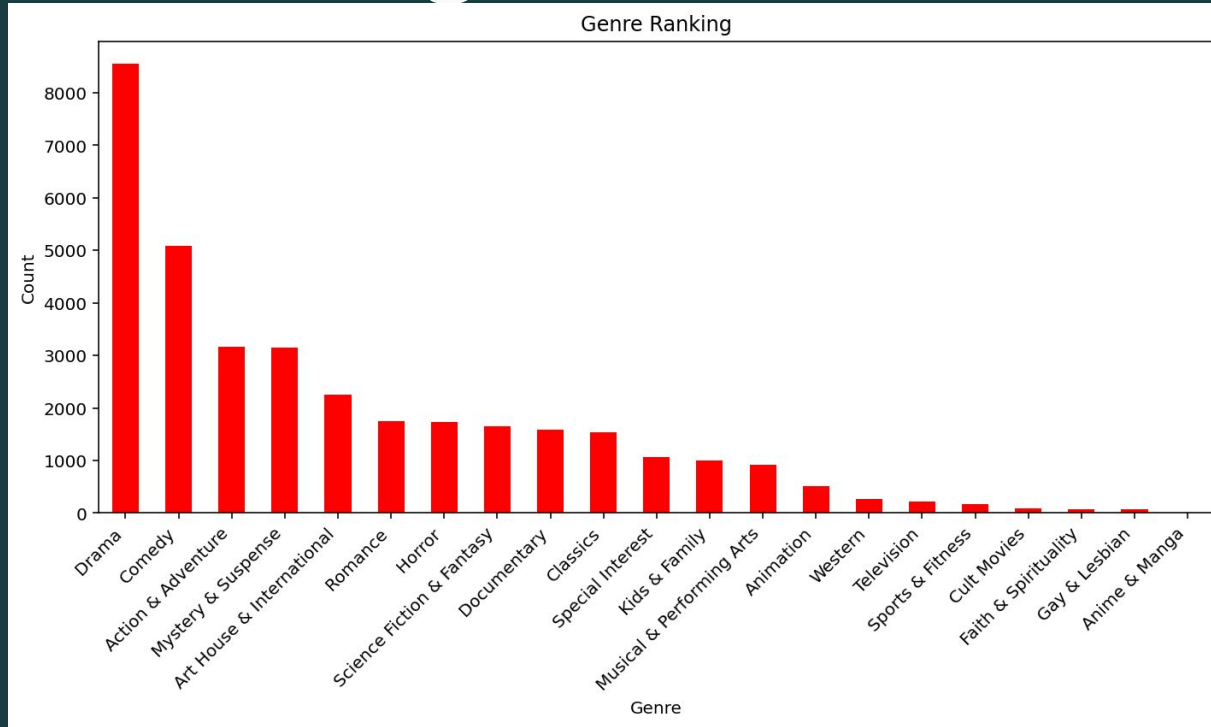
Exploratory Data Analysis

Genre Ranking - Movies



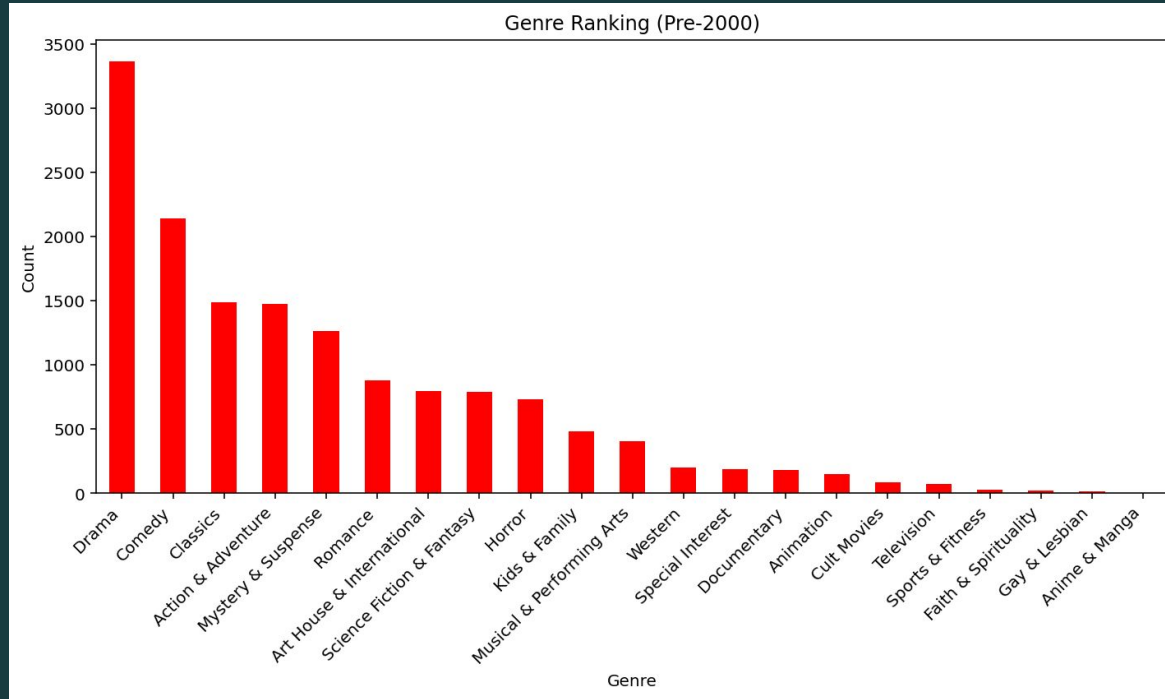
- Drama is the most common genre in the Movies dataset, followed by Action and Comedy
- Genres such as Western, Musical, and War were less frequently produced

Genre Ranking - Rotten Tomato



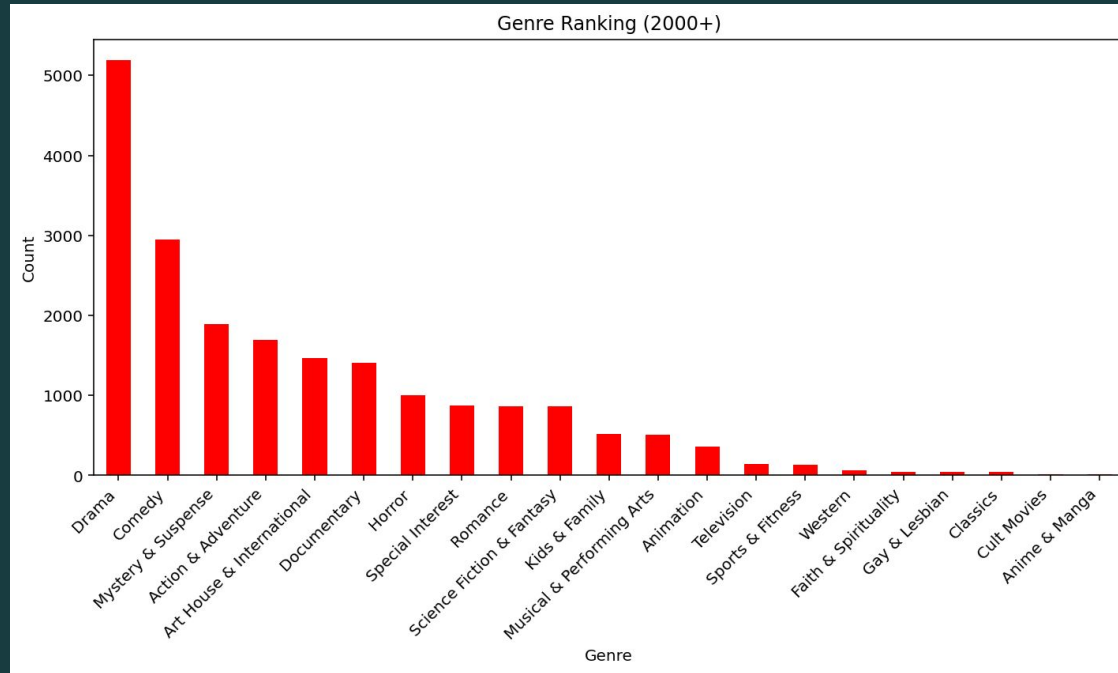
- Years range from 1914 to 2019
- Overall, Drama, Comedy, and Action & Adventure were the most common genres throughout the years.
- Movies less likely to be created is Anime & Manga.

Genre Ranking - Rotten Tomato



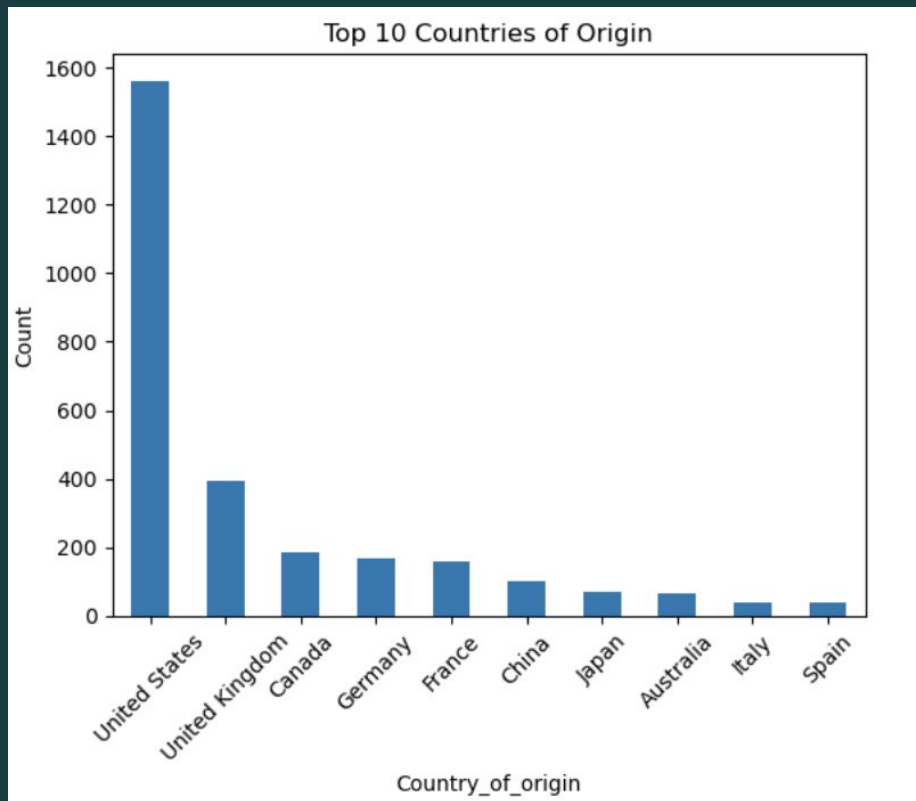
- Years range from 1914 to 2000
- Overall, Drama, Comedy, and Classics were the most common genres.
- Movies less likely to be created is Anime & Manga.

Genre Ranking - Rotten Tomato



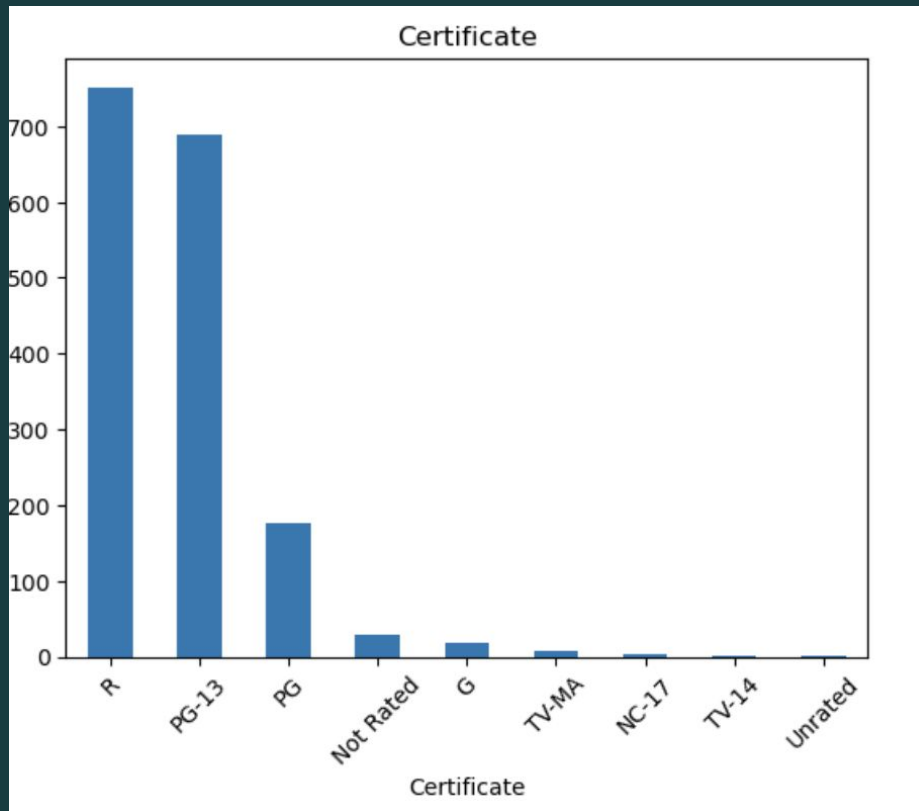
- Years range from 2001 – 2019
- Overall, Drama, Comedy, and Mystery & Suspense were the most common genres.
- Movies less likely to be created is Anime & Manga.

Top 10 Countries



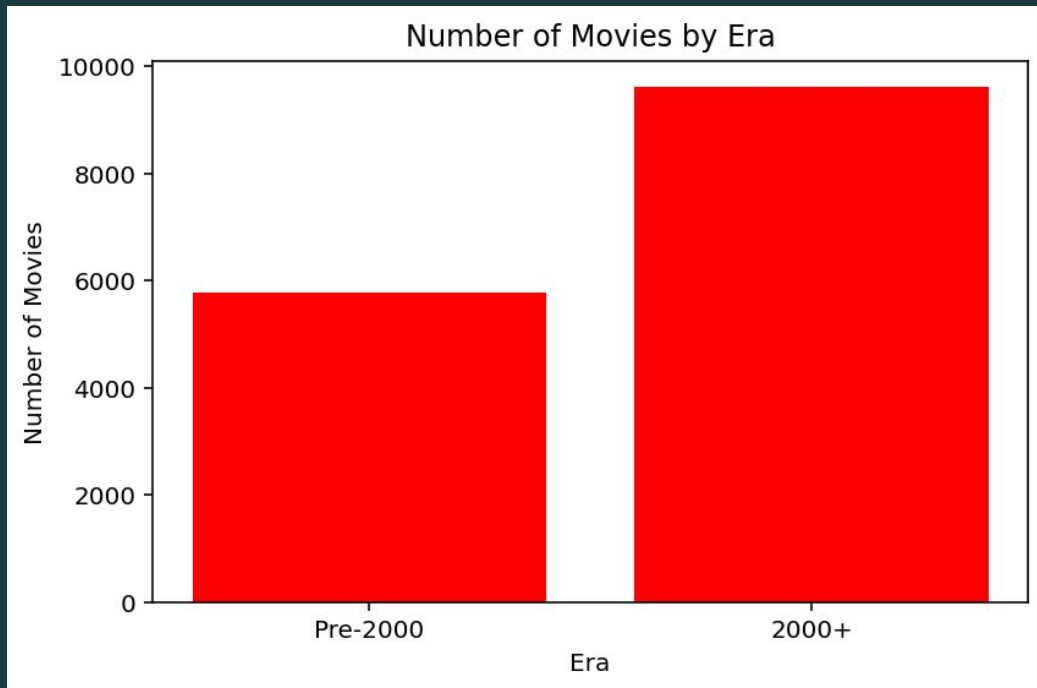
- This graph shows the top 10 countries of origin for movies in the dataset -> with the United States producing the majority of movies then followed by the United Kingdom
- Countries such as Italy and Spain have contributed the least amount

Movie Certificate Ranking



- R-rated movies are the most common
- PG-13 movies are the second most common
- PG-rated movies are the third most common
- Ratings like G, TV-MA, NC-17, TV-14, and Unrated are less frequent
- There are also some Not Rated movies, but far fewer compared to R and PG-13

Movies by Era

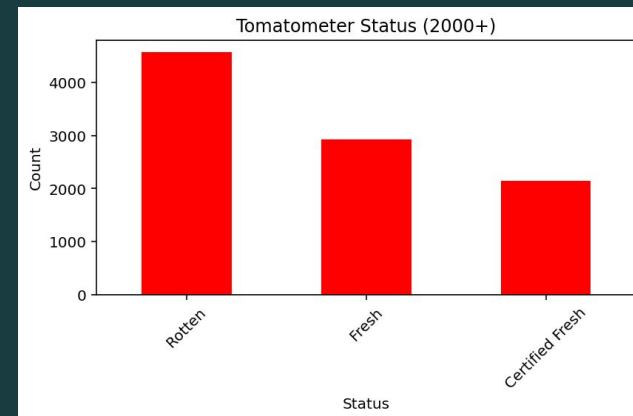
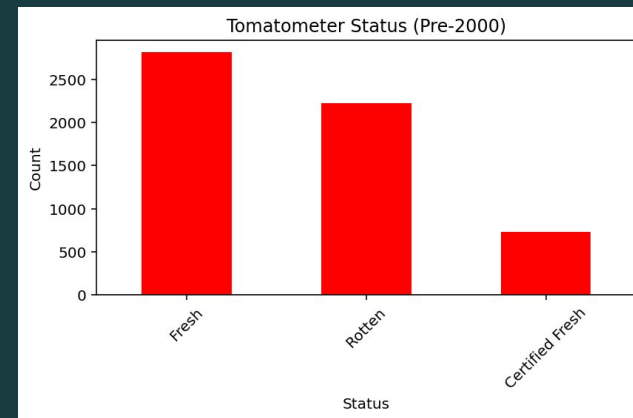
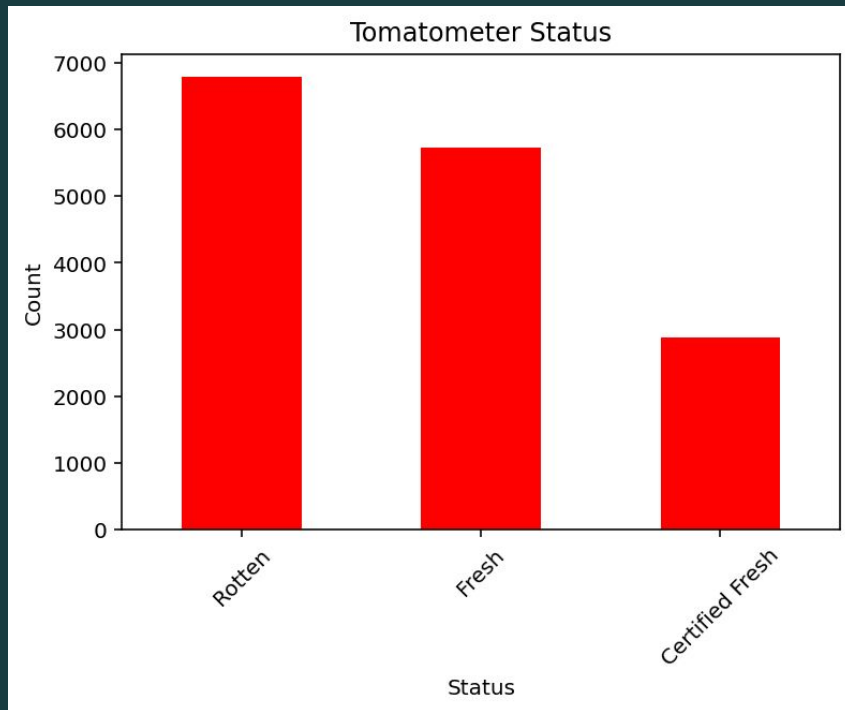


Out[157]:

	era	year	movie_count
84	Pre-2000	2000	304
202	2000+	2014	722

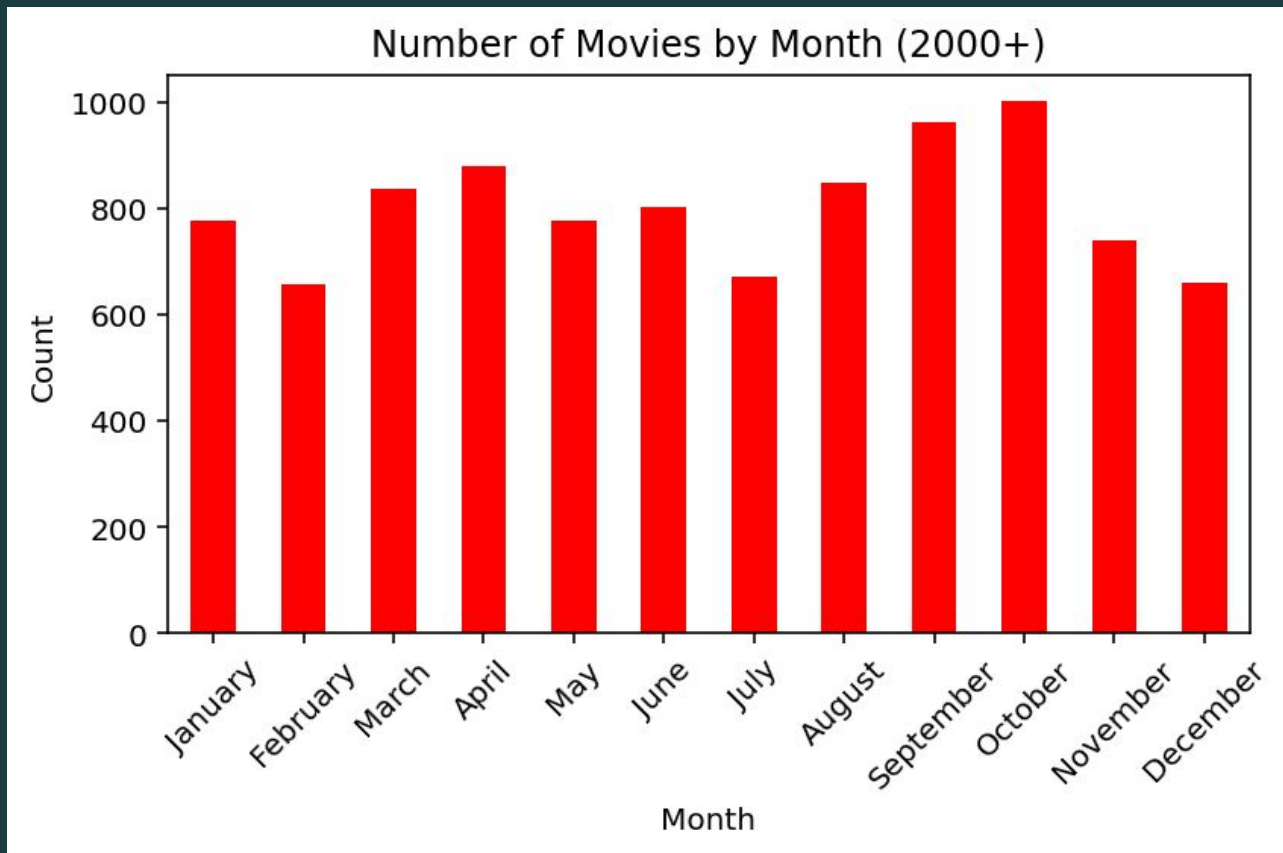
- Pre-2000 : 5774
- 2000+ : 9622
- The highest year for movies was 2000 and 2014
- $P(2000) = 5.3\%$
- $P(2014) = 7.5\%$

Tomatometer Status



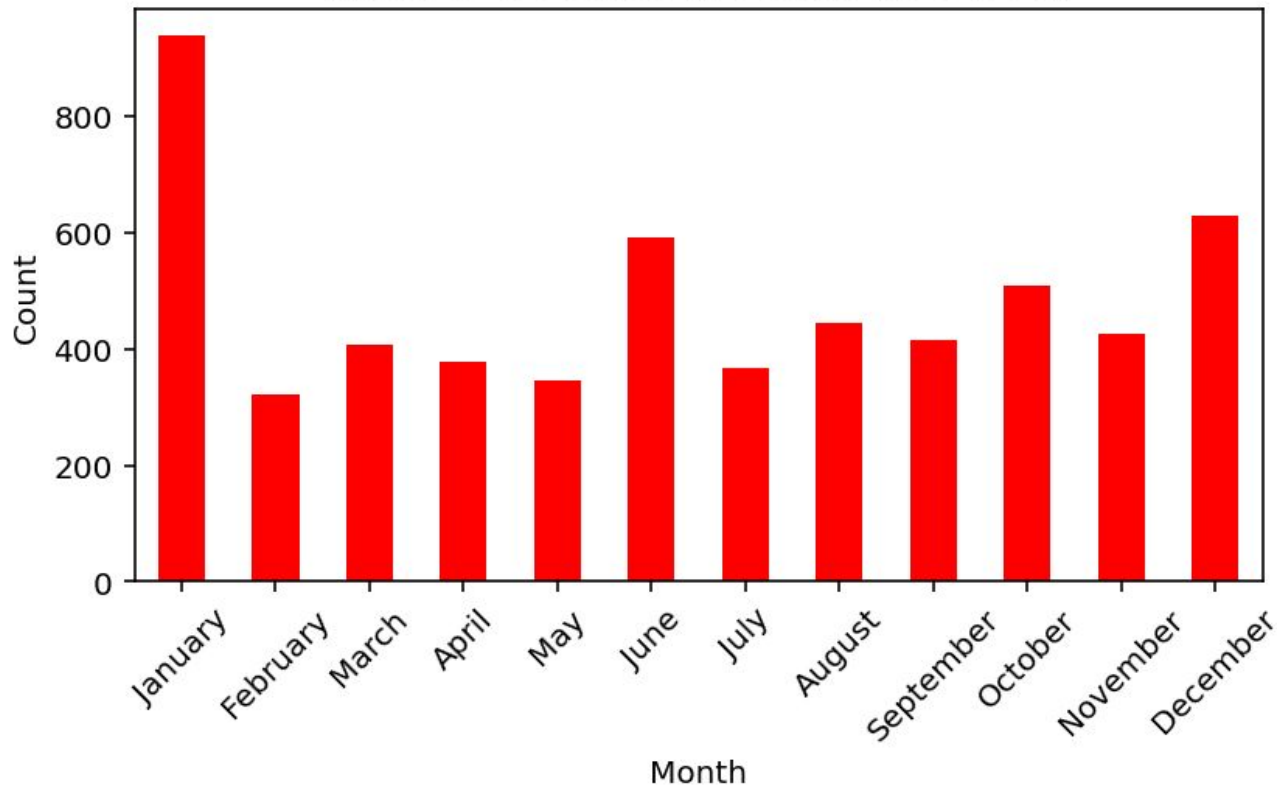
- Demonstrated how rating has changed and progressively grow more criticism for a better "Fresh" status

Movies by Month

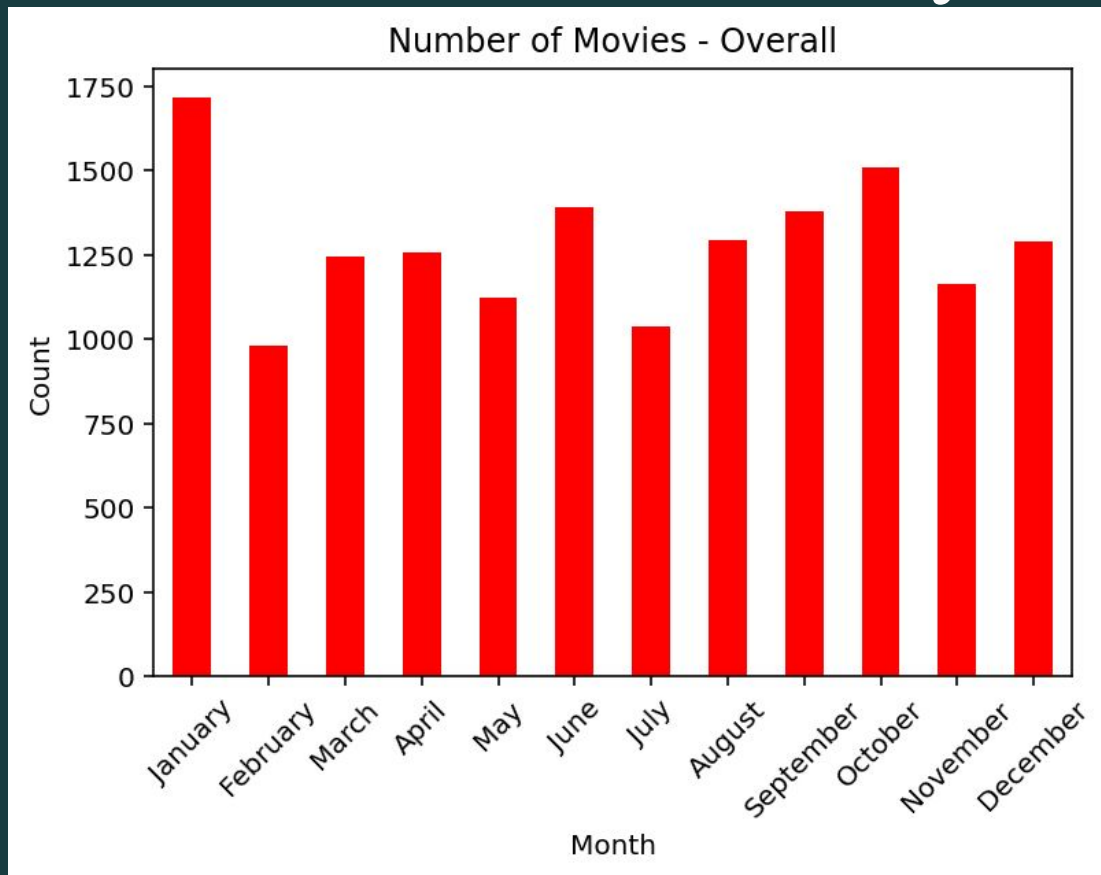


Movies by Month

Number of Movies by Month (Pre-2000)

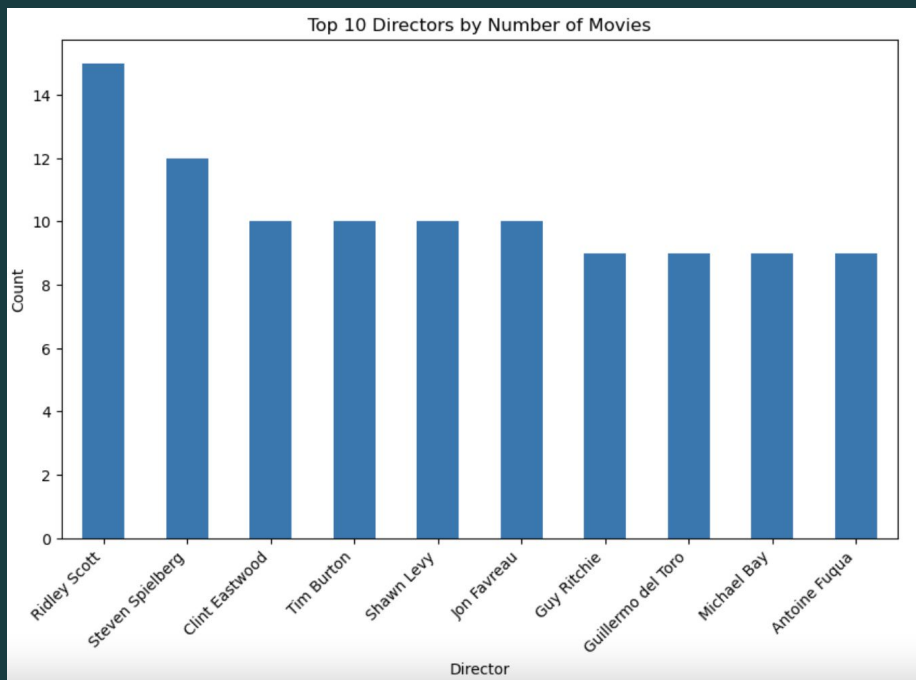


Movies by Month



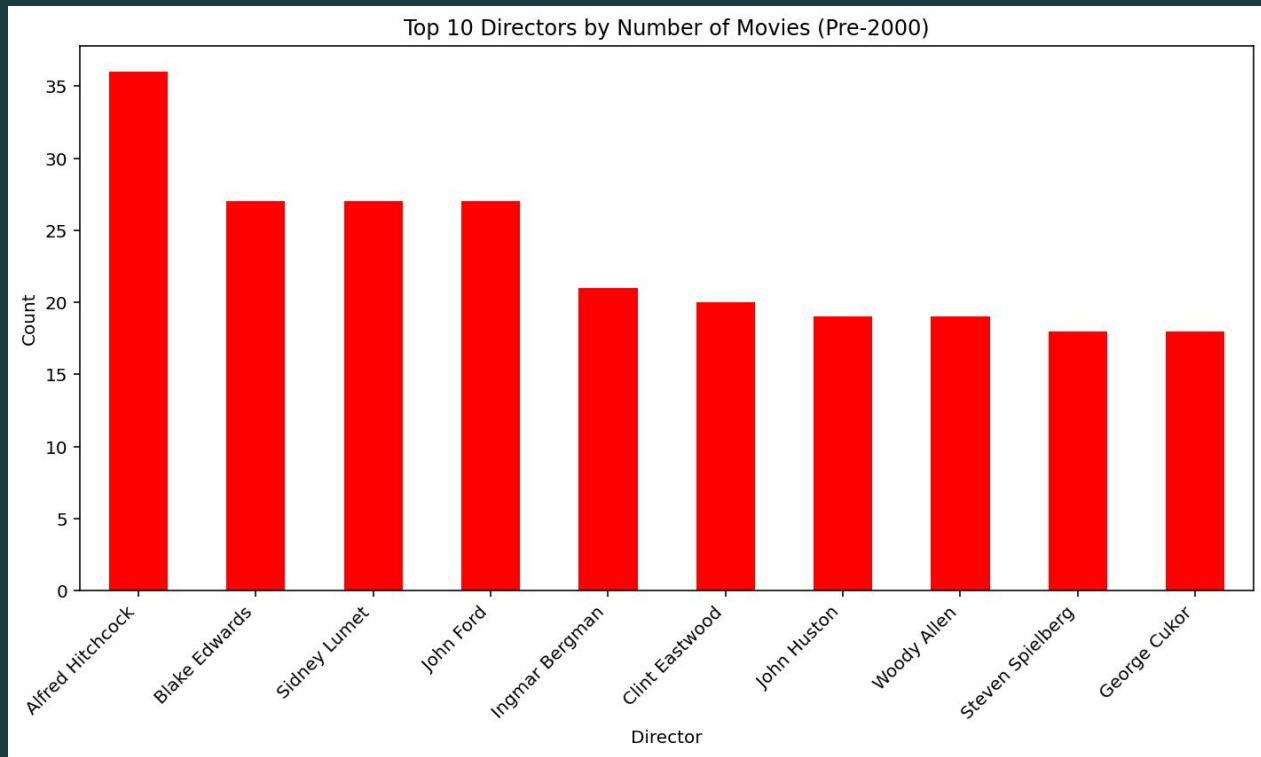
```
Out[185]:  
season  
Winter      25.9%  
Spring      23.6%  
Summer      24.2%  
Fall        26.3%  
Name: count, dtype: object
```


Top 10 Directors by Number of Movies

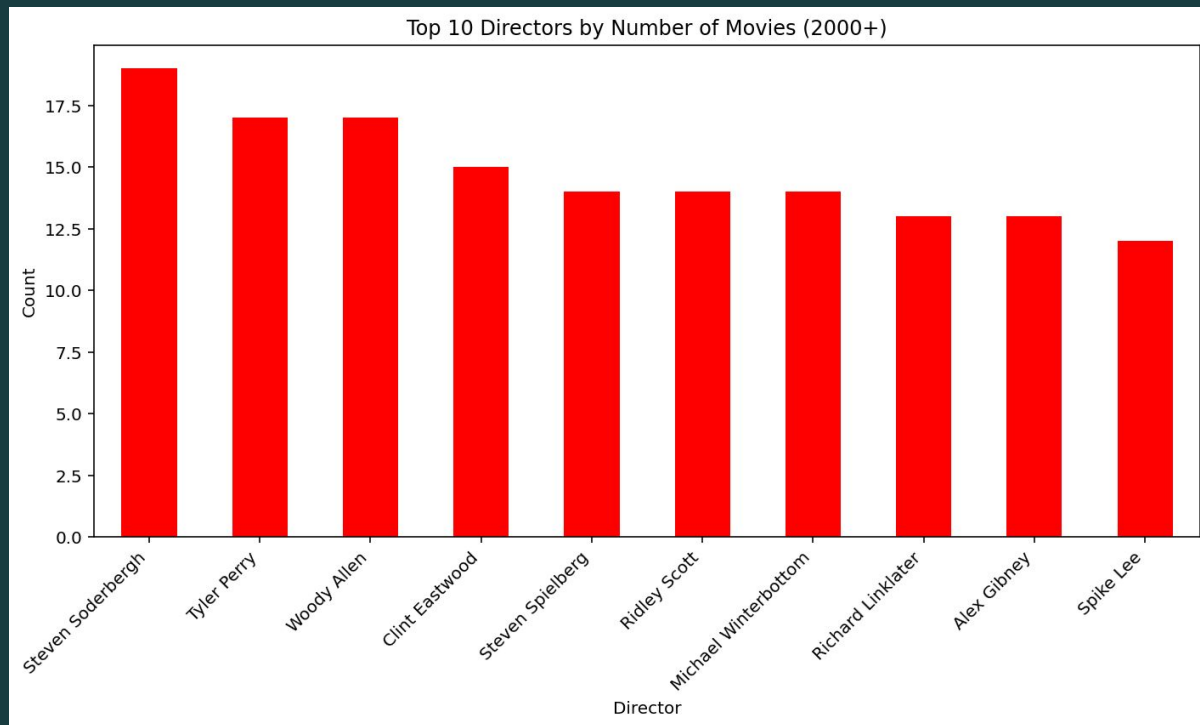


- Ridley Scott has directed the most movies
- Steven Spielberg is the second most frequent director
- Directors Clint Eastwood, Tim Burton, Shawn Levy, and Jon Favreau have slightly fewer movies but are still among the top contributors with similar numbers of movies

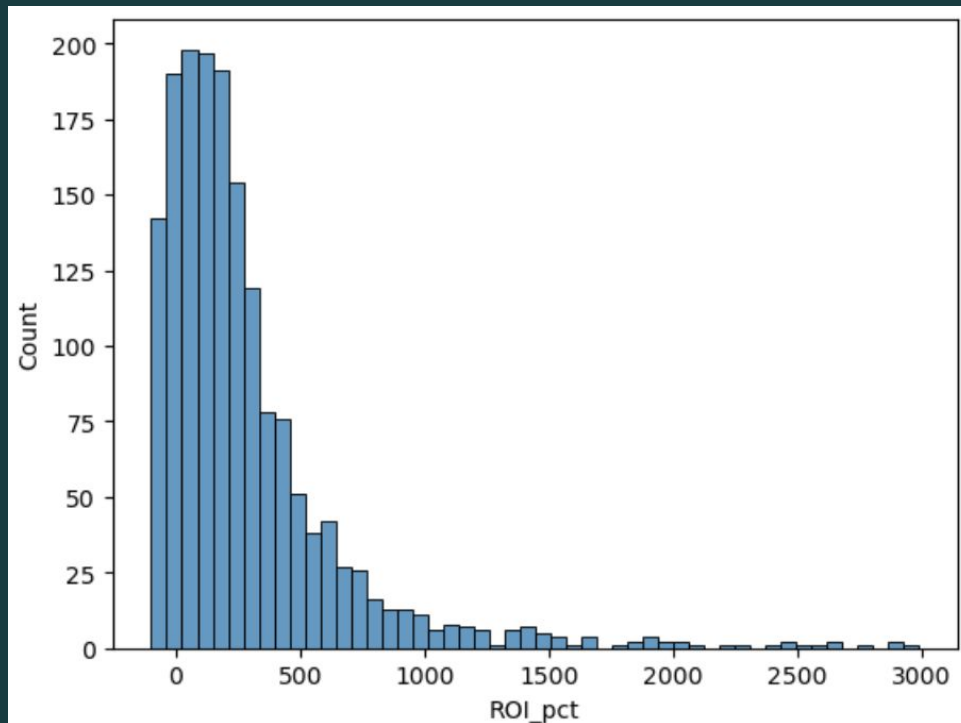
Top 10 Directors by Number of Movies



Top 10 Directors by Number of Movies



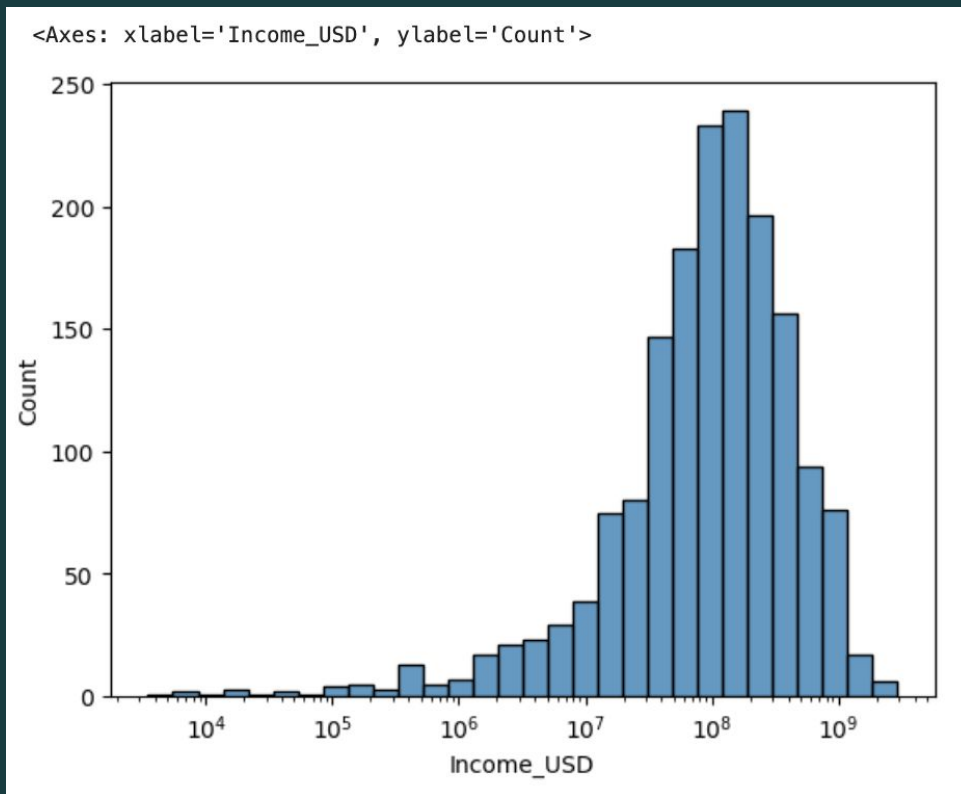
Return on Investment



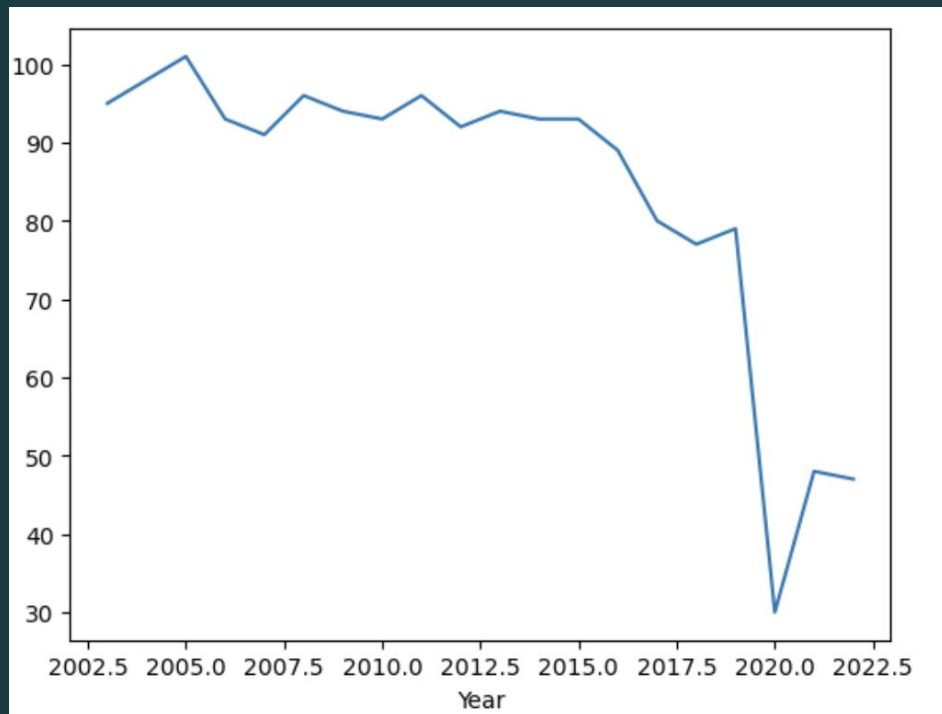
- Most movies have relatively low to moderate returns on investments while a small number of films get relatively high return on investment
- The distribution is right skewed showing that there are a few highly profitable outliers

Income Distribution

- Most films earn within a common range but the movie industry's profits shown to be dominated by a small number of huge successes

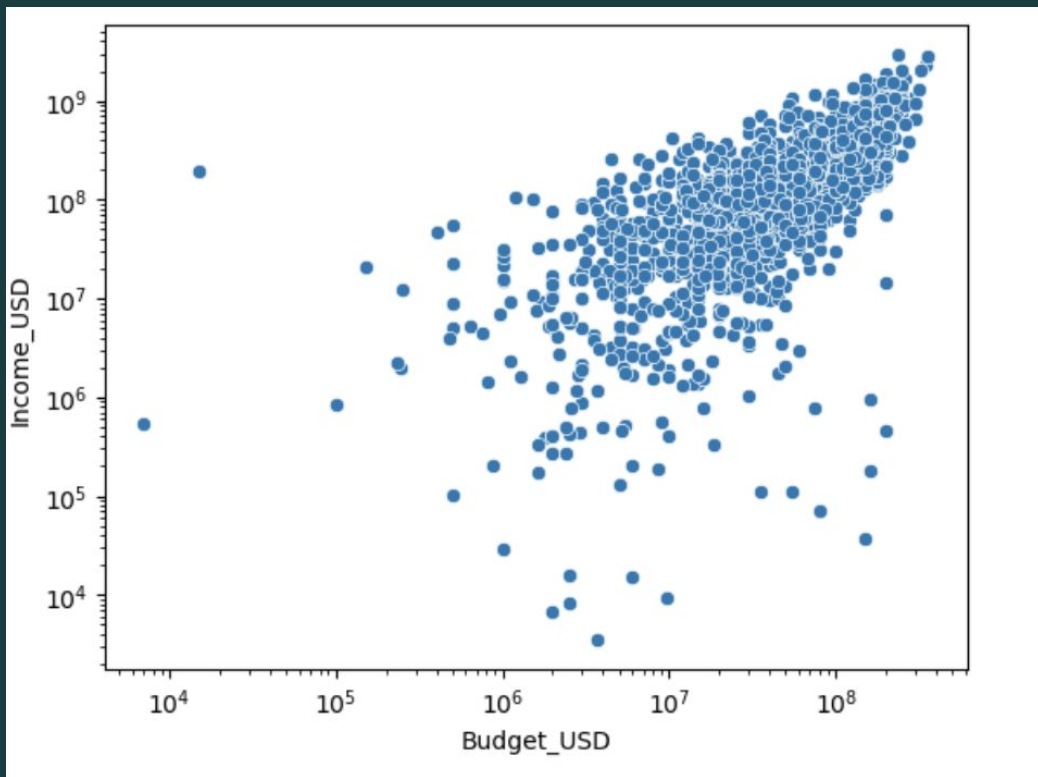


Number of Movies by Year



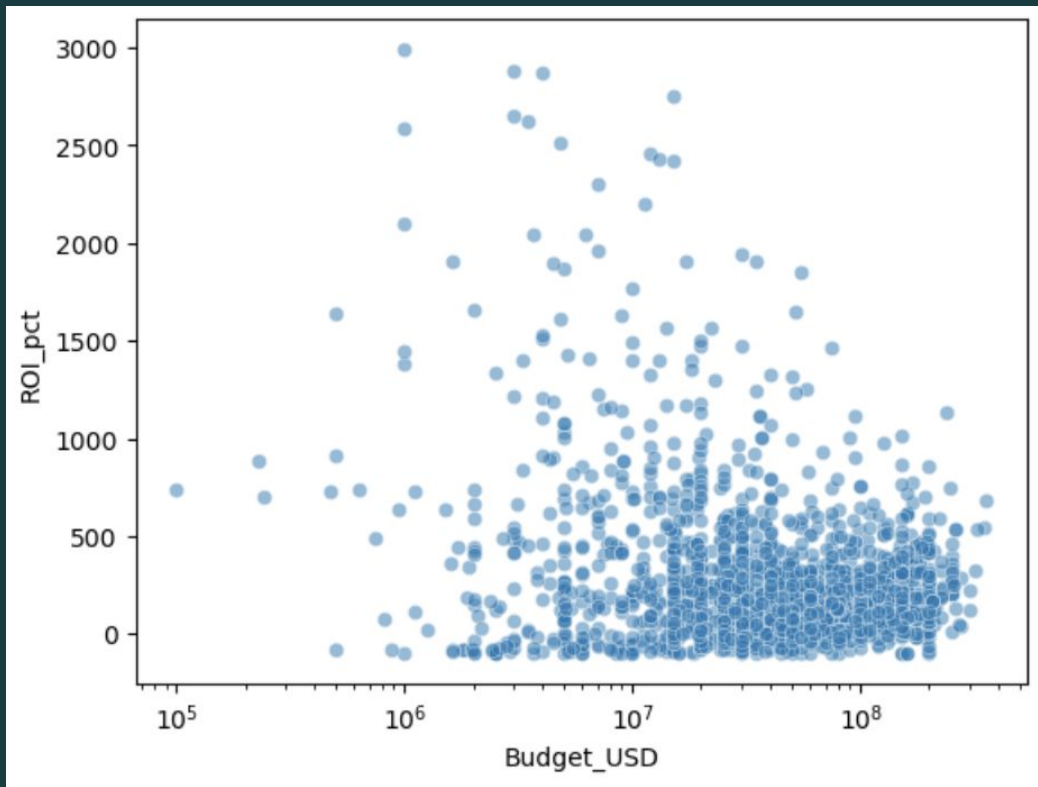
- There's long term stability in the 2000s and early 2010s
- There's a gradual decline in the industry due to COVID 19 which eventually leads to a partial recovery

Income and Budget Correlation



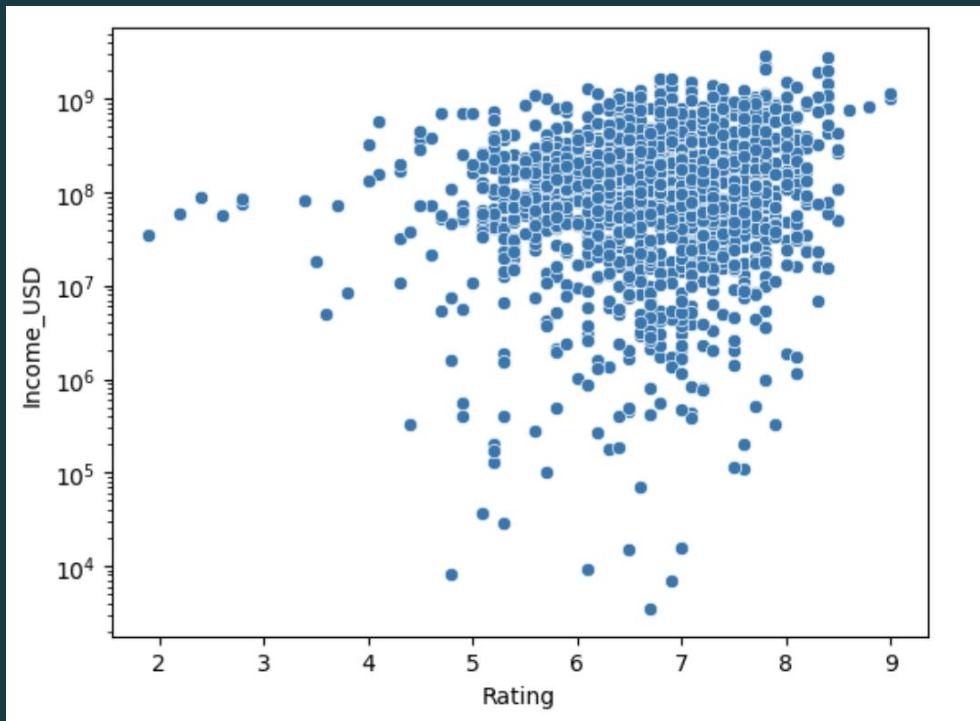
- There's a strong positive correlation between having a high movie budget and the income that is produced from it
- There are still considerable variability and several outliers

Movie Budget and ROI Correlation



- Bigger budgets can be believed to reduce uncertainty but it does not necessarily increase profitability
- Lower-budget movies can outperform movies in return of investments

Income and Rating Correlation

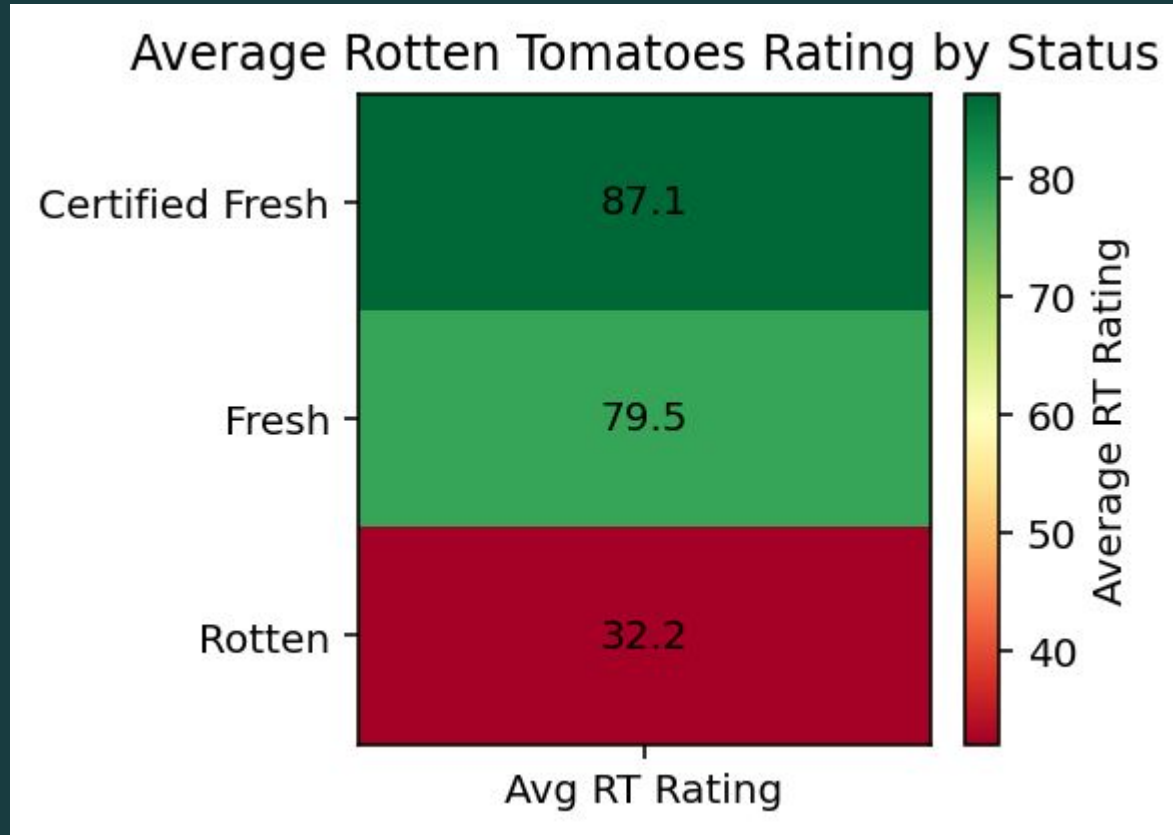


- There is no strong correlation between high ratings and high income
- However, the data suggests a slight positive relationship between the two

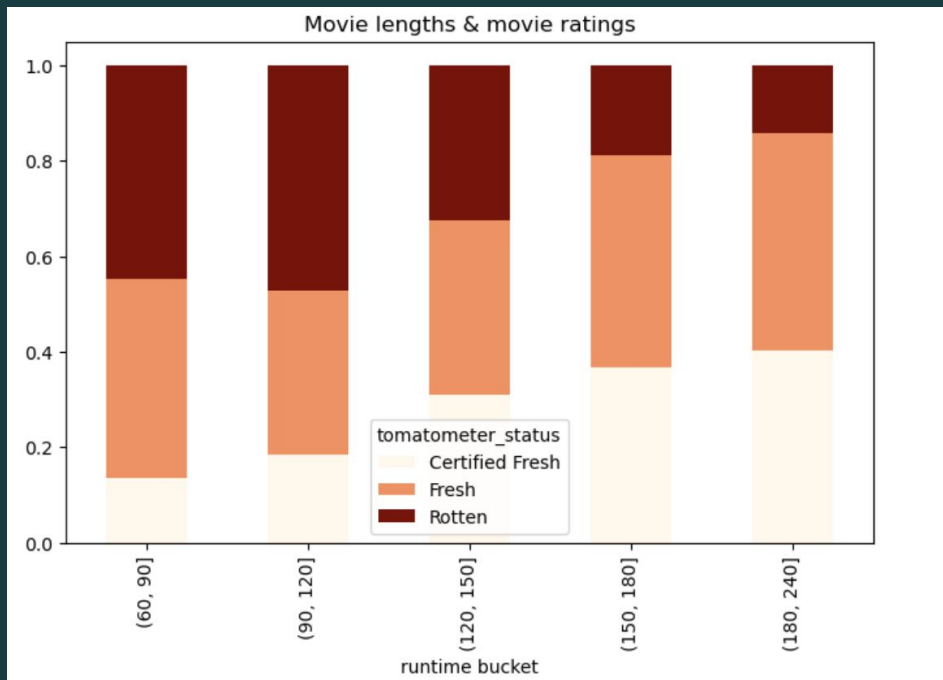
Average Movie Ratings by Genre

	mean_rating	count
Genre		
War	7.500000	17
Biography	7.370175	114
History	7.222727	44
Western	7.080000	10
Sport	7.007692	26
Drama	7.000756	794
Animation	6.994286	70
Music	6.943590	39
Crime	6.841727	278
Musical	6.786667	15
Mystery	6.770492	183
Thriller	6.679667	300
Romance	6.620705	227
Action	6.600972	617
Sci-Fi	6.598571	210
Adventure	6.573975	488
Comedy	6.510721	541
Family	6.393269	104
Fantasy	6.378125	192
Horror	6.307583	211

Correlation between Rating and Status



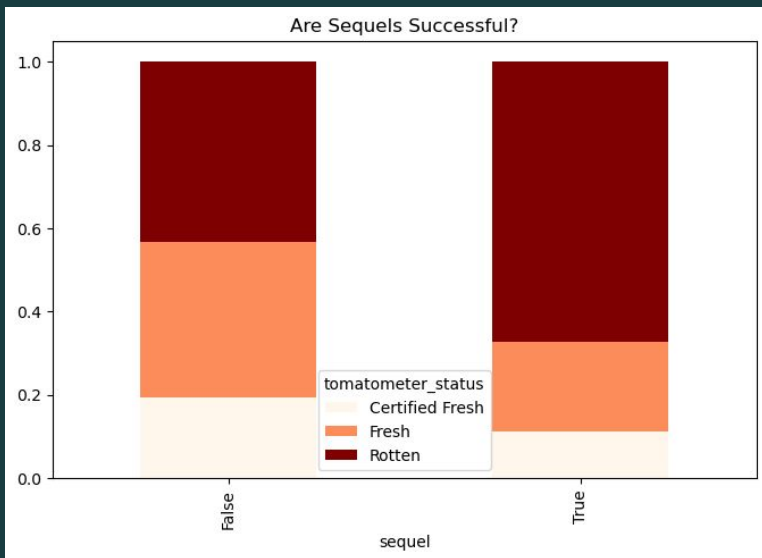
Runtime Minutes



- Although most movies are from 90–120 minutes, the movies that ran longer had less rotten ratings

```
runtime bucket
(90, 120]      9375
(60, 90]       3659
(120, 150]     1720
(150, 180]      255
(180, 240]       77
Name: count, dtype: int64
```

Franchise movies



Sequential Movies that were good

	movie_title	year
1151	8 1/2	1963
1818	The Godfather, Part II	1974
2035	Star Wars: Episode IV - A New Hope	1977
2371	Superman II	1981
2464	Star Trek II: The Wrath of Khan	1982
2475	Mad Max 2: The Road Warrior	1982
2644	Star Trek III - The Search for Spock	1984
2842	Star Trek IV - The Voyage Home	1986
3043	Evil Dead 2: Dead by Dawn	1987
3395	Back to the Future Part III	1990
3601	Terminator 2: Judgment Day	1991
4087	The Legend of Drunken Master (Jui kuen II) (Dr...	1994
4345	Richard III	1995
4816	Scream 2	1997
5104	Toy Story 2	1999
6215	Stuart Little 2	2002
6322	Spy Kids 2: The Island of Lost Dreams	2002
6380	Amandal A Revolution in Four-Part Harmony	2003
6708	Kill Bill: Volume 2	2004
6732	3-Iron	2004
6834	Spider-Man 2	2004
7036	Shrek 2	2004
7232	Star Wars: Episode III - Revenge of the Sith	2005
8308	3:10 to Yuma	2007
8347	2 Days in Paris	2007
8489	4 Months, 3 Weeks and 2 Days	2007
8681	Che: Part Two (Guerrilla)	2008
8942	Hellboy II: The Golden Army (Hellboy 2)	2008
9602	Piranha 3-D	2010
9646	Toy Story 3	2010

Sequential Movies that were Rotten

	movie_title	year
1809	Phase IV	1974
2022	Exorcist II: The Heretic	1977
2120	Damien---Omen II	1978
2213	Zombi 2	1980
2221	Oh, God! Book II	1980
2237	Smokey and the Bandit II	1980
2305	Friday the 13th Part 2	1981
2314	Halloween II	1981
2382	Omen III: The Final Conflict	1981
2399	Piranha Part Two: The Spawning	1981
2403	Grease 2	1982
2449	Airplane 2 - The Sequel	1982
2461	Halloween III: Season of the Witch	1982
2476	Death Wish II	1982
2479	Amityville II: The Possession	1982
2480	Friday the 13th Part 3	1982
2516	Jaws 3	1983
2579	Superman III	1983
2583	Psycho II	1983
2587	The Sting II	1983
2590	Friday the 13th - The Final Chapter	1984
2663	Cannonball Run II	1984
2705	Rambo: First Blood Part II	1985
2719	Friday the 13th, Part V - A New Beginning	1985
2724	A Nightmare on Elm Street 2: Freddy's Revenge	1985
2748	Howling II	1985
2749	The Hills Have Eyes, Part 2	1985
2762	Police Academy 2: Their First Assignment	1985
2783	Rocky IV	1985
2793	Death Wish 3	1985

- Not 100% accurate- sequential data was found if with any of the following
 - 2|3|4|I|II|III|IV|Part|part|Chapter|chapter
- Does not take into consideration movies that are not sequential but happen to contain the above occurrences (ex: 8 ½ movie)

For all Attempts

```
models = { "Decision Tree" : DecisionTreeClassifier(random_state=42),
           "Random Forest": RandomForestClassifier(n_estimators=500,
                                                    random_state=42),
           "Naive Bayes": BernoulliNB(),
           "Logistic Regression": LogisticRegression(max_iter=5000)
}

for types, model in models.items():
    print(types)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    print(classification_report(y_test, y_pred))

    conf = confusion_matrix(y_test, y_pred)
    plt.figure(figsize=(8,6))
    sns.heatmap(conf, annot=True, cmap="gray_r", fmt="d", cbar=False)
    plt.xlabel('predicted labels')
    plt.ylabel('true labels')
    plt.title('confusion matrix')
    plt.show()
```

- For loop to go through all the models and return accuracy, precision, recall etc
- Added a confusion matrix to each loop

First try -Imbalanced- most columns

Checking for imbalanced data

```
rt['tomatometer_status'].value_counts()

##certified fresh is a lower than the other 2, might oversample

tomatometer_status
Rotten          6691
Fresh           5624
Certified Fresh  2884
Name: count, dtype: int64
```

Initially Kept

- Rating- one hot encoded
- Month- one hot encoded
- years
- Directors- one hot encoded
- Studio name- one hot encoded
- Tomatometer_count
- Audience_count

Dropped

- Genre- multiple listed values in one cell

First try - Results

Naive Bayes

	precision	recall	f1-score	support
Certified Fresh	0.45	0.26	0.33	692
Fresh	0.61	0.35	0.44	1428
Rotten	0.53	0.81	0.64	1680
accuracy			0.54	3800
macro avg	0.53	0.47	0.47	3800
weighted avg	0.55	0.54	0.51	3800

- Issues with studio names and most likely director names are discrepancies/duplications on values that we would consider the same but have slight differences in spelling or specificity
- Genre was not considered

- Accuracy stayed in the 50s for all models (not much better than flipping coin at random)
- One hot encoding increased column # by number of unique occurrences of directors, studio names

```
'studio_name_Fox Home Entertainment',  
'studio_name_Fox International',  
'studio_name_Fox Lorber',  
'studio_name_Fox Meadow Films',  
'studio_name_Fox Network',  
'studio_name_Fox Searchlight',  
'studio_name_Fox Searchlight Pictures',  
'studio_name_Fox Searchlight Pictures, Film4 and Waypoint',  
'studio_name_Fox Walden',  
'studio_name_Fragile Films',  
'studio_name_Fragmighty',  
'studio_name_France 2 Cinéma',  
'studio_name_France 3 Cinéma',  
'studio_name_Francinex',  
'studio_name_Franco London Films',  
'studio_name_Freestlye Releasing',
```


Second try- balanced - few columns

...	Mystery & Suspense	Romance	Science Fiction & Fantasy	Special Interest	Sports & Fitness	Television	Western
...	1	0	0	0	0	0	0
...	0	0	0	0	0	0	0
...	0	0	0	1	0	0	0
...	0	0	0	0	0	0	0
...	0	0	1	0	0	0	0

Kept

- Rating- one hot encoded
- Month- one hot encoded
- Yrs_bin- bucketed years
- genre list- one hot encoded
- Tomatometer_count
- audience_count

Removed

- directors
- studio name

Machine Learning - feature engineering

```
### making title into id aka index
rt = rt_sorted.drop_duplicates(

    subset=['movie_title', 'year'],
    keep='first'
)

##bucket years into decades
print(rt['year'].min())
print(rt['year'].max())
yrs_bins = [1914, 1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020]
yrs_labels = ['early 1900s', '50s', '60s', '70s', '80s', '90s', '00s', '10s']

rt['yrs_bin'] = pd.cut(
    rt['year'],
    bins=yrs_bins,
    labels=yrs_labels
)

rt = rt.set_index(['movie_title', 'year'])

rt.head()
```

```
movie_title      year
Cabiria          1914  early 1900s
The Birth of a Nation  1915  early 1900s
A Fool There Was    1915  early 1900s
Intolerance        1916  early 1900s
Broken Blossoms     1919  early 1900s
...
In the Aisles (In den Gängen)  2019      10s
Missing Link                  2019      10s
5B                            2019      10s
Level 16                      2019      10s
The LEGO Movie 2: The Second Part  2019      10s
Name: yrs_bin, Length: 15199, dtype: category
Categories (8, object): ['early 1900s' < '50s' < '60s' <
```

Second try- results

Decision Tree

	precision	recall	f1-score	support
Certified Fresh	0.64	0.62	0.63	692
Fresh	0.59	0.60	0.59	1428
Rotten	0.61	0.61	0.61	1680
accuracy			0.61	3800
macro avg	0.61	0.61	0.61	3800
weighted avg	0.61	0.61	0.61	3800

Random Forest

	precision	recall	f1-score	support
Certified Fresh	0.65	0.78	0.71	692
Fresh	0.76	0.65	0.70	1428
Rotten	0.70	0.73	0.71	1680
accuracy			0.71	3800
macro avg	0.70	0.72	0.71	3800
weighted avg	0.71	0.71	0.71	3800

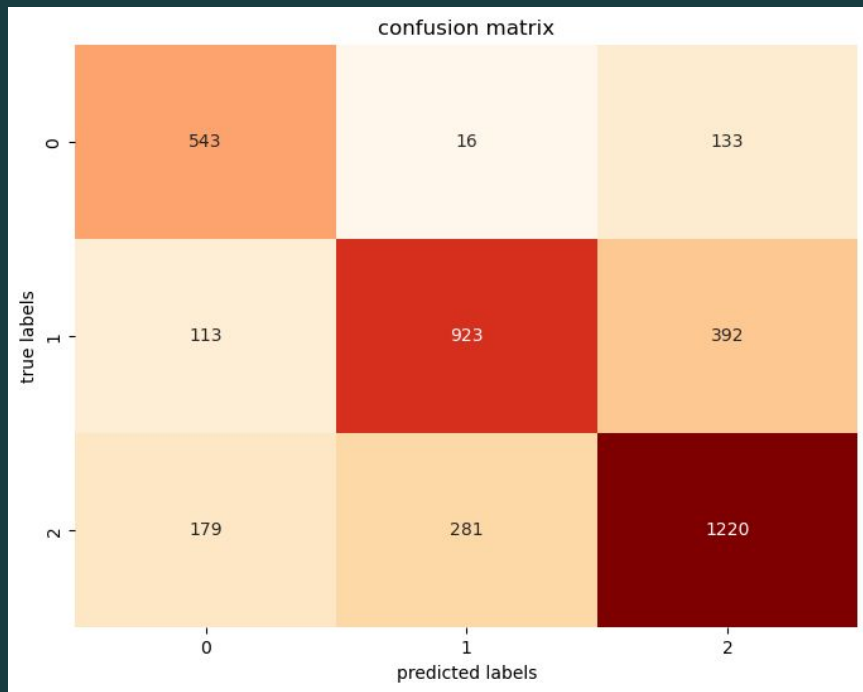
Naive Bayes

	precision	recall	f1-score	support
Certified Fresh	0.30	0.33	0.31	692
Fresh	0.65	0.57	0.61	1428
Rotten	0.62	0.65	0.64	1680
accuracy			0.56	3800
macro avg	0.52	0.52	0.52	3800
weighted avg	0.57	0.56	0.57	3800

Logistic Regression

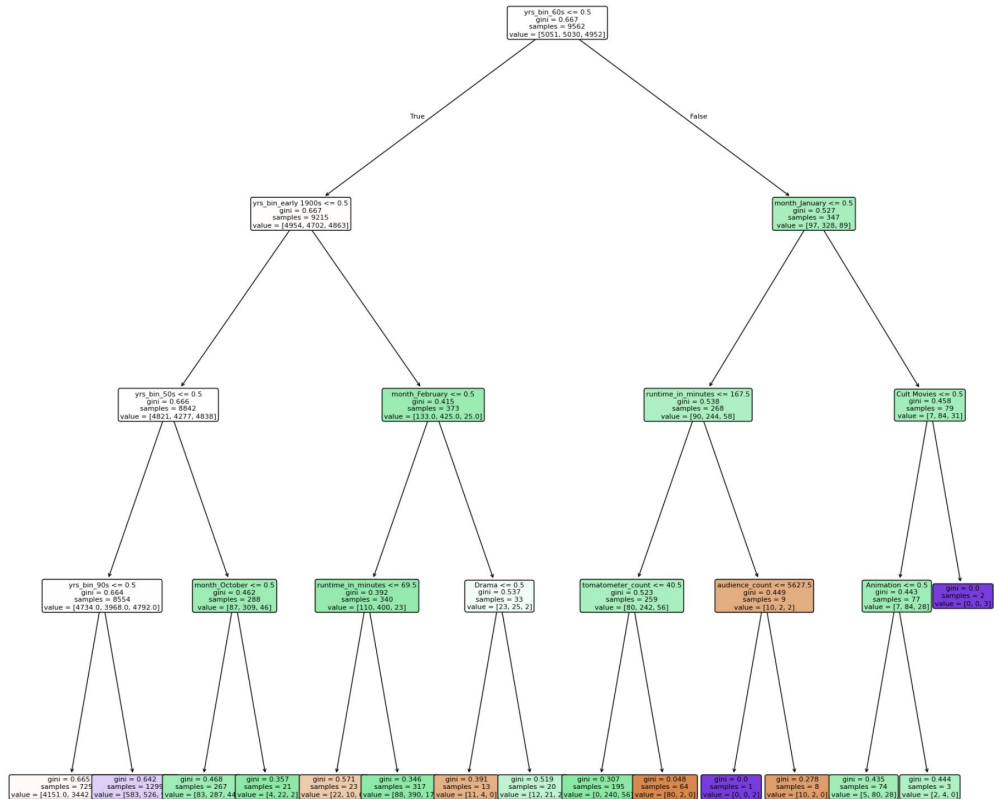
	precision	recall	f1-score	support
Certified Fresh	0.54	0.64	0.59	692
Fresh	0.68	0.64	0.66	1428
Rotten	0.69	0.67	0.68	1680
accuracy			0.66	3800
macro avg	0.64	0.65	0.64	3800
weighted avg	0.66	0.66	0.66	3800

Second try- Random Forest



- Most errors occur between target values that are close to each other
- It's unlikely to classify a certified fresh as rotten and vice versa, compared to confusing a fresh with certified fresh

Second try- Random Forest



- An example of one tree in forest
- Release era being a root = strong feature
- Runtime is also a strong indicator

Third try- add columns

```
good_movies = y_train.isin(['Certified Fresh', 'Fresh'])
X_train_directors = rt.loc[X_train.index, 'directors']
director_counts = good_movies.groupby(X_train_directors).sum()
```

```
def director_bin(num):
    if (num >= 20):
        return 'very_popular'
    elif (num >= 10):
        return 'popular'
    elif (num >= 5):
        return 'known'
    elif (num >= 1):
        return 'emerging'
    else:
        return 'who?'
```

```
director_label = director_counts.apply(director_bin)
```

movie_title	year	directors	director_label
War Horse	2011	Steven Spielberg	very_popular
Ready Player One	2018	Steven Spielberg	very_popular
Lincoln	2012	Steven Spielberg	very_popular
Bridge of Spies	2015	Steven Spielberg	very_popular
The Post	2018	Steven Spielberg	very_popular

Kept

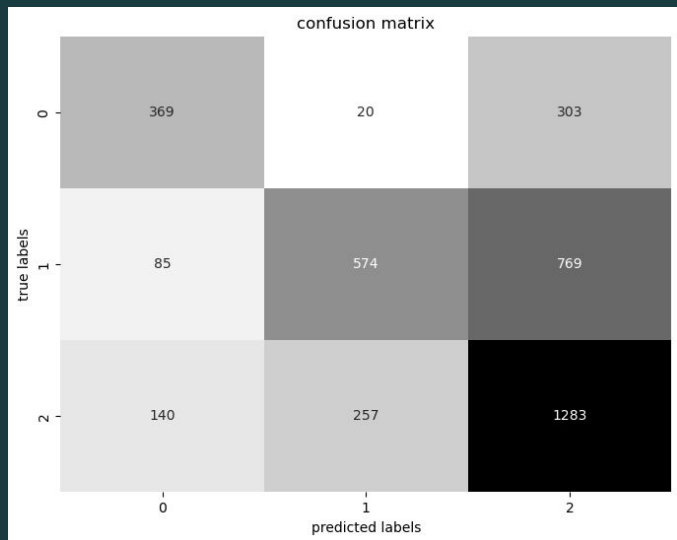
- Rating- one hot encoded
- Month- one hot encoded
- Yrs_bin- bucketed years
- genre list- one hot encoded
- Tomatometer_count
- Audience_count
- Directors- bucketed

Removed

- studio name

Third try- Results

Random Forest				
	precision	recall	f1-score	support
Certified Fresh	0.62	0.53	0.57	692
Fresh	0.67	0.40	0.50	1428
Rotten	0.54	0.76	0.64	1680
accuracy			0.59	3800
macro avg	0.61	0.57	0.57	3800
weighted avg	0.61	0.59	0.57	3800



Learnings:

- Director repudiation is not a strong predictor if a movie will be ranked rotten or otherwise
- Doesnt consider how many good movies a director made/ how many movies director made total

Hypothesis Testing

1. Our assumption is that Thrillers grossed highly in the early 2000s, while Superheros dominated in the later years
2. Franchise movies like Final Destination and Marvel movies had a higher chance of being in the top 100
3. Movies released during summer were more successful.
4. Movies featuring at least one well-known directors were more successful.
5. Ratings have progressively become more criticized based on the increased volume of movies being created.
6. Movies with a runtime slightly over two hours were more successful.

Learnings

- Movie success is not based on one factor
- High investments in a movie suggests higher return through income
 - Although lower investment in movies doesn't always guarantee lower income
- Rating and revenue have a weak relationship and can't fully support a conclusion
 - High ROI doesn't mean high rating
- Director reputation alone is a weak predictor
- A movie is likely to be a Drama and R rated
 - Filmed in the USA during the Winter season
- Using matplotlib to create clear visualizations made trends and patterns easier to find and interpret

To Consider (future improvements)

- Finding a pairing dataset with enough rows to have more meaningful features
- Cleaning '*Studio Names*' so they are more generalized
 - Using fuzzy search to match similarly named studios
- Normalizing runtime minutes? Or bucketing them in model
- Improving genre
 - Focus on multiple genre groups (if under romance and comedy)
- Instead of directors, give a score based on #good movies/total movies

Thank you